From the Wall to Resolution Geometry

Why Coherence Was Not Enough

Green Response Kernels, Recursive Modulation, Rupture, and the Reconstruction of an Observable Geometry

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Abstract

Resolution Geometry (RG) did not begin with an observational quotient. Its motivating observation was a delayed return after a wall tap, together with changes of perceived localization under repetition. In the author's recollection, the first mathematical home of that observation was a Green response kernel: a direct way to write an impulse, propagation, and a later response. The subsequent CTMT programme enlarged this description into Recursive Modulation Impulse (RMI), spectral kernel assembly, regularized inversion, recursive prediction, and the Terror Kernel's treatment of rupture and survival. These constructions made the observation calculable but did not establish that a stable reconstruction was uniquely determined by the record.

This paper reconstructs the intervening conceptual bridges from the supplied archive rather than projecting mature RG backwards onto childhood experience. It follows sensing maps, uncertainty, spatial and temporal compression, phase-selective rigidity, Fisher rank, null projection, CRSC stability diagnostics, information-geometric trigonometry, resolved/unresolved coupling, saturation, exact law equivalence, gauge completeness, projectable transport, causal cones, and physical calibration. Each bridge is distinguished from the stronger claim it could not justify. The decisive reversal is from preserving a coherent latent explanation to identifying what every observationally equivalent explanation must share. Coherence remains a legitimate property of specified signals or states; it loses its role as a universal warrant for reality, identifiability, or geometry.

The reconstruction includes a source and code audit, explicit counterexamples, and fourteen reproducible mathematical witnesses. It is an intellectual and mathematical genealogy, not a dated priority history, a retrospective acoustic dataset, or a proof that all physical geometry follows from delay alone.

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1 How to read this history

Three kinds of statement must not be confused.

1. **Authorial recollection.** The author identifies the wall-tap experience as an early motivation, recalls a childhood belief in coherence, and states that its first mathematical expression used a Green kernel because that was the easiest available representation. These are retrospective statements supplied for this account, not conclusions inferred from file timestamps.

2. **Documentary reconstruction.** The CTMT HTML archive and the supplied RMI–Terror extract contain the formulas, protocols, and Python demonstrations discussed below. Later RG manuscripts document the increasingly explicit distinctions among observation, inversion, invariance, transport, and physical interpretation.

3. **Mathematical reconstruction.** Some transitions are explained here by counterexamples and propositions. These show why a bridge is logically needed; they do not prove the precise order in which every idea occurred to the author.

The order of exposition is therefore a dependency history. No undocumented childhood age, date, instrument precision, or sequence of manuscript revisions is assigned. The supplied HTML is a composite archive, not an untouched earliest notebook. Its present contents cannot independently certify the first written occurrence of a formula. The claim about the Green kernel's priority within the author's development is attributed to the author, while the archive corroborates the presence of that response-kernel programme [1, 2, 3].

The aim is neither to vindicate every CTMT assertion nor to dismiss the intermediate work as disposable speculation. The operational parts already contain real inverse-problem machinery. Conversely, successful code execution cannot repair a dimensional contradiction or prove a uniqueness theorem. The history becomes clearer when these two facts are retained together.

The central change. The early question was approximately: *What coherent process can produce and preserve this return?* The mature question is: *Which distinctions are justified by the declared observation law, and what additional assumptions are needed for geometry?* The latter is not merely a more elaborate answer to the former. It changes what counts as an answer.

2 The wall: an event, an interval, and a return

The autobiographical account describes rhythmic tapping at different walls, remembered return delays, changes of loudness, and a sudden change in perceived echo localization as the rhythm accelerated. The archive associates this transition with an early, informal "Terror Calculus" of coherence rupture and re-synchronization [2, 4, 5]. The defensible historical datum is the experience of ordered events and changing response, not a recovered calibrated timing record.

Write only

$$e_{\text{emit}} \prec e_{\text{receive}}.$$

A later calibrated description may introduce $\Delta t = t_1 - t_0$, but this already presupposes a clock convention and a way to assign the two events. Source onset, propagation, wall vibration, transduction, detection, and recollection need not have been separately resolved. A perceived localization transition is likewise not direct evidence that the room's physical geometry changed.

2.1 Why coherence was an attractive interpretation

The author's recollection that coherence was believed in as a child is taken literally. The following explanation of its appeal is retrospective interpretation, not an invented psychological diagnosis. Repetition offers a natural contrast: a return can seem to hold together as a recognizable event, or to become ambiguous when responses overlap. A word such as coherence can collect regularity, persistence, synchrony, recognizable location, and successful expectation into one intuitive picture.

That picture gives a productive research question: what preserves a response across changes of rhythm, wall, or observation? Its weakness is that several different jobs are being performed by one word. Phase agreement, low variance, stable perception, faithful prediction, and unique identification are not equivalent. The later work was required precisely because the initial intuition did not distinguish them.

Nothing in this history requires a child to have formulated Fisher information, a quotient, or a theory of causal cones. The mature mathematics is a reconstruction of the question, not a hidden theory attributed to the original experience.

3 Why the Green kernel came first

3.1 The shortest mathematical route from tap to response

For a declared linear causal response operator L, a Green function solves, schematically,

$$LG(\cdot, x') = \delta_{x'}, \qquad y = G * s.$$

For an ideal isolated delayed return, the simplest time-domain representation is

$$G(t) = A\delta(t - \tau), \qquad y(t) = As(t - \tau).$$

A broadened response replaces the delta distribution by a pulse or decay envelope. A repeated source $s_T(t) = \sum_n p(t - nT)$ produces

$$y_T(t) = \sum_n (G * p)(t - nT).$$

Thus delay, attenuation, overlap, and rhythm enter without a new ontology. This is a mathematical explanation of why the author's first representation was easy: it maps a localized excitation directly to a later response. These equations are explanatory normal forms here, not claimed verbatim transcriptions of a lost childhood derivation.

The archive's "Green Kernel (linear propagation)" explicitly assigns it the role of spectral propagator carrying phase and amplitude. Its "Green Kernel (Spectral)" protocol proceeds from impulse responses to Fourier spectra, basis expansion, regularized inversion, and comparison against time-delay profiles [2, 3]. The Green kernel is therefore a genuine documentary starting point, not a name attached after RG was finished.

3.2 What this representation already assumes

A Green response is not a derivation of its operator. It assumes a model class, source and receiver conventions, boundary conditions, a propagation law, and the regime in which linearity is adequate. It describes how a chosen system responds. It does not establish that only one system could have produced the response.

Even the spectral route has a specific information loss. For a time shift,

$$h_\tau(t) = h(t - \tau), \qquad \widehat{h}_\tau(\omega) = e^{-i\omega\tau}\widehat{h}(\omega),$$

so $|\widehat{h}_\tau|^2 = |\widehat{h}|^2$. Power spectra alone do not determine absolute delay. Phase, a trigger reference, or additional measurements must remain available. A reconstruction protocol that moves from full impulse traces to spectral power and then claims the original timing has to justify that passage.

3.3 Two different meanings of kernel

This distinction is indispensable to the whole history:

$$\underbrace{G(t,x;x')}_{\text{response kernel}} \qquad \neq \qquad \underbrace{X \times_F X}_{\text{kernel relation of an observation law}} \quad .$$

The first maps excitation to response. The second pairs descriptions that produce the same law. The birth of RG is not a proof that these two mathematical objects are identical. It is a change of focus from constructing the first to understanding the ambiguity of the second.

4 RMI: from one response to a family of operational kernels

4.1 Spectral assembly and phase normalization

The RMI archive proposes a constructive modulation principle from which different operational kernels are assembled. A representative normalized form is

$$K(x,x') = \int_{\Omega_\omega} M(\omega;\gamma,\Theta,Q,\phi,T) e^{i\Phi(x,x';\omega)/\mathcal{S}_*}\, d\omega. \tag{1}$$

The archive distinguishes the spectral domain, measure, envelope, phase/action, dimensional prefactor, and action scale. Stationary-phase and path-sum representations organize contributions under additional regularity and asymptotic assumptions [3].

This is more ambitious than the first Green description: a common assembly language is proposed for propagation, coherence envelopes, temporal projection, and other kernel families. The useful bridge is the explicit separation of envelope, phase, measure, and observation. The unproved stronger claim would be that writing many responses in one integral establishes a universal physical generator.

The archive sometimes prints $e^{i\Phi}$ after assigning action units to Φ, whereas surrounding definitions use $e^{i\Phi/\mathcal{S}_*}$. Those forms are compatible only if the phase has been redefined as dimensionless. The reconstruction records the convention issue instead of silently treating all occurrences as identical.

4.2 The observation operator is already present

The operational section writes

$$O_k = \iint L_k(x,x')K(x,x')\,dx\,dx' + \epsilon_k.$$

With a basis expansion $K \approx \sum_j \kappa_j b_j$, this becomes

$$O = A\kappa + \epsilon.$$

This is a substantial ancestor of RG: the kernel and its measured projection are different objects. A sensor family, noise law, and basis determine which combinations of the kernel are accessible. The later question of observational fibres is already latent in the possible null space of A, although the mature exact-law quotient is not yet established by this equation.

4.3 RIP: reconstruction is not identification

The Recursive Inversion Protocol uses a weighted regularized inverse of the form

$$\widehat{\kappa} = (A^\top C_\epsilon^{-1} A + \lambda R^\top R)^{-1} A^\top C_\epsilon^{-1} O.$$

It makes estimation computable and can stabilize an ill-conditioned fit. It does not turn the chosen penalty into information supplied by the experiment.

Proposition 4.1 (A unique optimizer can select an unresolved representative)**.** *For the observation $O = \kappa_1 + \kappa_2$, all pairs with the same sum have the same noiseless observation. A strictly convex regularizer can nevertheless return a unique estimate. Different regularizers can return different estimates while predicting essentially the same observation.*

Proof. The fibre is an affine line. A strictly convex objective has at most one minimizer on that line, but its minimizer depends on the penalty. The observation equation does not choose the penalty. □

This is one precise point at which the real hunt for RG begins in the dependency reconstruction: a stable answer is not enough. The question becomes whether the record determines the answer, or whether the method has supplied it.

4.4 Recursion, prediction, and information survival

The Minimal Kernel Predictor adds a discrepancy correction,

$$\widehat{O}_{t+1} = \mathcal{F}[\kappa_t] + \alpha\{O_t - \mathcal{F}[\kappa_t]\}.$$

Its proposed coherence-based gain and the ICM information-survival language attempt to make recursive prediction accountable to observations. The demonstrations, however, include placeholder coherence values and entropy surrogates that are not the same as the mutual-information definition in the prose [3].

The intermediate achievement is a feedback architecture. The remaining gap is an independently specified law for the gain, a calibrated uncertainty interpretation, and a test that the recursively selected kernel is identifiable rather than merely self-consistent.

5 The intermediate operational bridges

5.1 Primitive extraction, anchors, and constants

The RMI programme extracts envelope widths, decay rates, spectral peaks, and proposed hop lengths, then combines them under regime bridges. It also seeks agreement among action and speed anchors. This introduced a valuable discipline: a symbol must be connected to a measurement, units, a regime, and uncertainty [3].

The unresolved issue is circularity. If a speed, scale, or constant enters the forward model or unit convention, recovering it from that same construction is not an independent derivation. A measured ratio such as E/ν can anchor an action scale only with its measurement and frequency convention specified. Cycle and angular frequency cannot be interchanged without the corresponding 2π factor.

In mature RG, anchors remain, but their logical role is explicit: they reduce a scale fibre or choose a calibrated representative. They do not disappear into a claim of emergence.

5.2 FMC, TUCF, and composition

The archive's spatial compression, temporal uncertainty compression, and joint FMC–TUCF demonstrations address a real problem: a response is not only a pointwise amplitude. It is integrated over spatial channels and temporal windows. A temporal kernel carries delay and anchor substitution; the joint model tries to preserve uncertainty through successive projections [3].

These were necessary intermediate questions for this programme, but not uniquely necessary historical inventions for every possible route to RG. Their mathematical residue is the demand for composable observation maps, explicit loss under coarse-graining, and covariance propagation through each map. A successful composite calculation still needs a proof that the retained features preserve the task.

5.3 From a scalar score to a covariance geometry

A single coherence or compatibility score compresses several possible failures. Full covariance retains direction, correlation, and task dependence. For $T = T(O)$,

$$\Sigma_T \approx DT\,\Sigma_O\,DT^\top.$$

This formula does not require coherence as an ontological principle. It requires a differentiable task and a justified local uncertainty model. Shared errors can cancel in one task and remain as a floor in another. The later uncertainty manuscript develops this distinction as geometry rather than as a final error bar appended to a preferred answer [13, 19].

6 Terror: making failure part of the construction

6.1 Rupture as an operator, not a synonym for noise

The Terror section introduces multiplicative rupture and additive shock, schematically

$$T[K] = \Xi K + \eta,$$

and places rupture fields inside spectral assembly. Nested rupture composes as

$$T_2[T_1[K]] = \Xi_2(\Xi_1 K + \eta_1) + \eta_2.$$

Its role is to address the loss of the coherent approximation, deformation of ensembles, and instability of reconstructed response [3]. This extends the programme from "what persists?" to "what breaks, and how can the failure be diagnosed?"

The formula alone does not prove non-normality, non-invertibility, or a universal collapse mechanism. Those properties depend on the specified operator, distributions, and domain. The term Terror is retained here as a historical mathematical label, not a claim about an observer's psychological state or a new physical force.

6.2 The executable redundancy–rigidity mechanism

The substantial embedded Python constructs toy amplitudes, phases, and weights and computes complex observables

$$O_k = \sum_i w_i \Xi_{k,i} e^{i\Phi_{k,i}/S_*}.$$

Its rupture filter retains contributions satisfying a threshold on $\Xi_i \cos(\Phi_i/S_*)$ and renormalizes surviving weights. Its default forces one survivor if all contributions fail. Therefore a nonempty returned ensemble is not evidence that the physical coherence criterion survived.

Redundancy weights routes by survival and estimated variability. Rigidity applies a phase-selective damping factor,

$$O_{\mathrm{rig},k} = \sum_i w_i \Xi_{k,i} e^{-\lambda_{\mathrm{rig}} d_{2\pi}(\Phi_{k,i}/S_*)} e^{i\Phi_{k,i}/S_*}.$$

These are concrete computational mechanisms, not merely metaphors. Their limitation is equally concrete: filtering and weighting are chosen procedures. Their agreement cannot by itself establish that the retained latent interpretation is unique. Phase-selective rigidity here is not automatically the same concept as a later RG uniqueness theorem.

The deformation code also uses Cauchy shocks. Finite sample variance can be printed for such a sample, but an untruncated Cauchy law has no finite population variance. A finite-variance uncertainty certificate therefore needs a different model or an explicit truncation. The code's existence does not settle that modelling choice.

6.3 The lasting bridge: adversarial diagnosis

The durable inheritance is not that all rupture is one universal phenomenon. It is that a model must expose its own failure conditions. Null controls, hidden response changes, alternative paths, rank loss, and contradictory routes become scientifically relevant outputs. Mature RG retains this adversarial attitude while separating the distinct reasons an inference can fail.

7 CRSC, null projection, and the cost of identifying unlike nulls

The wider CTMT archive contains a further semi-step: an amplitude seed

$$\Psi_{\text{seed}} = \Xi e^{i\Phi/S_*}$$

is projected using

$$\Pi_{\text{null}} = I - HH^{+},$$

where H is called a Fisher curvature matrix. The CRSC construction combines coherence density with a modulation stability ratio involving oscillation, damping, and eigenvalues of selected blocks [2]. This is an attempt to replace an undifferentiated coherence judgment by directional stability.

Three distinctions are required before this can become RG geometry.

1. A Fisher matrix is a local information tensor. Its eigenvalues are not, merely by being called curvature, the Riemann curvature tensor of the associated metric.
2. $I - HH^{+}$ acts on the vector space on which H acts. Applying it to a scalar complex seed requires a specified embedding into that space. The displayed projection is not automatically well typed.
3. A statistical null direction is not a physical lightlike direction.

Proposition 7.1 (Statistical degeneracy and causal nullity are different)**.** *For a positive-semidefinite Fisher form I, a vector with $I(v,v) = 0$ lies in its radical. For a nondegenerate Lorentzian form g, a nonzero vector can satisfy $g(v,v) = 0$ while $g(v,\cdot) \neq 0$. Hence a Fisher kernel cannot be identified with a Lorentzian null cone merely by using the word null.*

Proof. Diagonalize the positive-semidefinite form: zero quadratic value forces all components in positive eigenspaces to vanish. For $g = \operatorname{diag}(1,-1)$, $v = (1,1)$ has zero quadratic value but $gv = (1,-1) \neq 0$. □

The projection–rupture pipeline has an additional implementation lesson. Its least-squares routine returns a matrix with shape controls × features, but then forms $J^{\top}J$ and interprets it as a control-space Fisher matrix. With that stored orientation, the control-space Gram matrix is $JJ^{\top}$. Both products are meaningful, but they live in different spaces. Mature RG must keep latent vertical directions and observation-space normal directions separate.

The same typing discipline applies to the archive's acceleration section. If v has units m/s, then dv/dS is not an acceleration unless S is a physical time parameter or a conversion to one is supplied. The printed proxy $|\nabla^2\Phi|/|\nabla\Phi|$ has inverse-length units for spatial derivatives, not the stated dimensionless units. These are failures of a proposed bridge, not evidence that no useful response model exists.

8 Information-geometric trigonometry: a real bridge, not the endpoint

The trigonometry manuscripts replace an assumed spatial distance by a statistical length,

$$d_F(\gamma) = \int \sqrt{I_{ij}(\theta)\dot{\theta}^i\dot{\theta}^j}\, d\lambda, \qquad I_{ij} = \mathbb{E}_\theta[\partial_i \log p\, \partial_j \log p].$$

This is a major change of subject: geometry is now tied to distinguishability of a declared observation model rather than to a single impulse amplitude. The older CTMT trigonometry text also proposes stationary-phase scaling and a coherence-saturated classical limit [6, 7].

What survives is the model-based statistical geometry. What does not follow is that flat Fisher geometry is already physical Euclidean geometry in metres. In a Gaussian location experiment,

$$Y \sim N(x, \sigma^2), \qquad ds_F^2 = dx^2/\sigma^2.$$

For every fixed σ this one-dimensional metric is flat, yet the Fisher length between fixed physical endpoints is $|x_1 - x_0|/\sigma$. Changing noise changes distinguishability without moving the endpoints. Flatness does not supply a ruler, a clock, a medium speed, or a causal cone.

Nor does writing a Fisher tensor prove it is the uniquely selected metric under every admissible rule. Selection requires a stated class and axioms. A geometry that descends to observable states and one uniquely characterized among all candidates are different achievements. This distinction becomes decisive in the repaired RG ladder [18].

9 Resolved and unresolved sectors: memory becomes a conditional theorem

The origin and elimination manuscripts give a precise replacement for informal re-projection. For a declared linear dynamical model,

$$\begin{aligned} \dot{x}_R &= A_{RR}x_R + A_{RN}x_N, \\ \dot{x}_N &= A_{NR}x_R + A_{NN}x_N, \end{aligned}$$

variation of constants yields

$$\begin{aligned} \dot{x}_R(t) = A_{RR}x_R(t) &+ A_{RN}e^{A_{NN}t}x_N(0) \\ &+ \int_0^t A_{RN}e^{A_{NN}(t-s)}A_{NR}x_R(s)\,ds. \end{aligned}$$

Thus the reduced memory kernel is

$$K_{\mathrm{mem}}(\tau) = A_{RN}e^{A_{NN}\tau}A_{NR}.$$

This is a genuine derivation once the dynamics and split are declared. It does not require an additional universal coherence field [4, 8].

The covariance counterpart is different. In a jointly Gaussian chart,

$$S_R = \Sigma_R - C_{RN}\Sigma_N^{-1}C_{NR}$$

is a conditional covariance. Correlated driving can produce $C_{RN} \neq 0$ even when the dynamic feedback kernel vanishes. One-way coupling can also create dependence without a round-trip feedback loop. Statistical seepage, dynamical memory, and geometric holonomy therefore require different evidence.

The old intuition that a return carries a trace of what was not directly observed survives here in a smaller form. Its content is not "everything reprojects because coherence exists." Its content is the displayed elimination identity under a specified model, together with null controls that distinguish coupling types.

10 The decisive reversal: exact observational equivalence

Let X be a candidate state space and let

$$F : X \longrightarrow \mathcal{P}(\mathcal{Y})$$

assign the joint observation law of the declared experiment. Define

$$x \sim_F x' \iff F(x) = F(x'), \qquad Q_{\text{obs}} = X/\!\sim_F.$$

The full kernel relation is

$$\mathcal{K}_{\text{obs}} = X \times_F X \rightrightarrows X.$$

Now the central object is not the preferred coherent representative but the entire class of descriptions that the experiment cannot distinguish [18, 11, 16].

Theorem 10.1 (Observable tasks are exactly fibre-constant tasks)**.** *A function $T : X \to Z$ factors uniquely through $\pi : X \to Q_{\text{obs}}$ if and only if $F(x) = F(x')$ implies $T(x) = T(x')$.*

Proof. Define $\bar{T}([x]) = T(x)$. Fibre constancy makes this well defined; surjectivity of π gives uniqueness. Conversely a factorization is constant on every fibre. □

This elementary theorem changes the burden of proof. The question is no longer whether one reconstructed room, field, or kernel fits. It is whether the requested quantity is the same for all states producing the same law. If not, the correct output is a correspondence, bound, or unresolved fibre.

10.1 Why tolerance cannot define the exact quotient

A finite-resolution rule $d(F(x), F(x')) \le \eta$ is generally not transitive. With threshold one, 0 is close to .75 and .75 to 1.5, but 0 is not close to 1.5. The repaired presentation therefore separates exact law equality from confidence sets and decision tolerances [16, 18].

For a confidence set $C(Y)$ in law space, use the full preimage

$$\Theta_C(Y) = \{x : F(x) \in C(Y)\}.$$

Its task image preserves unresolved branches rather than hiding them behind a single fitted answer. A numerical threshold is a decision parameter, not a new ontological identity.

10.2 Rank is local; the fibre is global

A Jacobian can be full rank near each of two states that have the same observation. The map $x \mapsto x^2$ has this property at $x = \pm 2$. Conversely $x \mapsto x^3$ is globally injective although its derivative vanishes at zero. Rank, global separation, and finite-precision conditioning are distinct tests. No single coherence score can replace all three.

11 Gauge completeness, saturation, and presentation

11.1 Known symmetries need not generate every ambiguity

Suppose a declared groupoid $\mathcal{A}$ acts by transformations preserving F. Then

$$\mathcal{A} \cdot x \subseteq [x]_F.$$

Equality is an additional completeness claim. A gain-only representation group does not automatically include a hidden response filter, an unresolved label permutation, or an unmeasured medium alternative. Soundness says the declared transformations are silent. Completeness says every silent alternative has been accounted for [11, 9].

This is another correction to the coherence picture. A response can remain stable under every tested transformation while an untested alternative remains observationally indistinguishable. Stability under known changes is not exhaustive identification.

11.2 Saturation is relative to a declared task and class

For a family of observables $\mathcal{O}$, saturation asks whether their joint values separate the relevant states or orbits within the declared class. Delays can constrain geometry while leaving material response unresolved. Additional amplitudes, spectra, or independent channels can refine the fibres. A Fisher matrix of selected features does not automatically certify that the full observation law has been retained [10, 5].

Finite examples cannot prove separation against every imaginable latent alternative. Approximate saturation needs an explicit metric, resolution, candidate class, and a bound controlling the alternatives. A finite battery tests the declared witnesses; it does not establish a uniform forgetting or completeness theorem merely by failing to find a counterexample.

11.3 Presentation independence is narrower than latent reconstruction

Two latent descriptions can expose the same law image without having identical internal variables or a canonical map between their hidden states. A quantity defined directly on that image is presentation independent in the corresponding sense. A latent lift or reconstruction is a stronger demand [16, 9].

The mature programme thus distinguishes three jobs that early coherence language tended to merge: preserve an observable, classify all observable distinctions, and reconstruct a hidden realization. Success at the first does not imply success at the other two.

12 Transport, holonomy, and the missing connection

A quotient is not yet a dynamics. Let Φ_t be latent evolution. It induces an observable evolution only if

$$F(x) = F(x') \implies F(\Phi_t x) = F(\Phi_t x')$$

for the relevant times. This is projectability: observationally equivalent starting descriptions must remain equivalent after the declared evolution. On a regular stratum, the observable tangent is the quotient

$$T_x X / \ker DF_x,$$

not a uniquely selected horizontal subspace of $T_x X$ [11, 12].

Proposition 12.1 (The law does not generally select a latent lift)**.** *For $F(x, z) = x$, the transformations $(x, z) \mapsto (x, z + ax)$ preserve F but carry the lift $(1, 0)$ to $(1, a)$. Hence the observation law alone cannot select a lift invariant under all these vertical shears.*

Proof. The first coordinate is unchanged, while the derivative sends $(1, 0)$ to $(1, a)$. Different a give different complements to the vertical direction. □

This no-go applies when both observable and unresolved directions are present and the full local shear freedom is admitted. It is not a claim that every possible connection is impossible. A declared positive latent covariance P can select, for a full-row-rank linearized observation J, the lift

$$B_P = PJ^\top (JPJ^\top)^{-1}.$$

Its canonicity is relative to P and the declared class, not supplied by the quotient alone [12].

Holonomy likewise requires a connection or specified transport rule and a closed-loop comparison. A loop discrepancy is not derived from covariance alone. This is where the mature geometry becomes richer than a delay calculation while also becoming more conditional than a universal coherence ontology.

13 From information geometry to causal and physical geometry

13.1 A causal cone needs causal input

An admissibility cone can encode permitted observable motions under a declared dynamics and uncertainty model. A physical characteristic cone requires additional propagation structure. To descend a latent cone to the quotient, its projected value must be constant along complete observation fibres. In real dimension at least three, a nondegenerate Lorentzian quadratic cone determines its quadratic form up to a nonzero scalar; a signature/time convention restricts the allowed sign, but does not fix the positive scale [11, 14, 18].

No Fisher degeneracy theorem alone supplies this cone. No equality of laws alone fixes a spacetime signature. No cone alone fixes a clock, a rod, or a universal propagation constant. These are precisely the bridges that the early identification of null, soft, collapse-free, and invariant-speed directions had compressed too aggressively.

13.2 The physical-distance bridge returns to the wall

For a declared homogeneous monostatic path,

$$z = b + a\tau + \epsilon, \qquad \tau = 2d/c_m, \qquad z = b + qd + \epsilon, \quad q = 2a/c_m.$$

Two distinct independent range anchors identify b, q in this affine model. They do not separately identify clock rate a and medium speed c_m. A calibrated a permits a duration report; a declared medium model permits the appropriate path conversion. A bistatic return is a path sum, not automatically twice a normal wall distance [15, 19].

For four labelled first-order returns in a known rectangle,

$$x = \tfrac{1}{2}(d_L + W - d_R), \qquad y = \tfrac{1}{2}(d_B + H - d_T).$$

Opposite-wall sums provide closure checks. Swapping left and right preserves those sums but changes x to $W - x$. Closure, labels, scale, and uncertainty therefore remain separate obligations.

This is the mature answer to the first Green kernel: the response model is retained, but its inverse is now accompanied by a statement of what is identifiable, what is anchored, and what remains ambiguous.

13.3 Consilience is not a new name for coherence

The crossing-into-physics manuscript compares routes with different assumption sets and preserves incompatible clusters rather than forcing one answer [17]. Its scientific role is agreement under independently vulnerable routes, not phase locking or subjective unity.

Even route agreement has a limit. If estimates share a common bias, their differences do not reveal it. With correlated uncertainties,

$$\mathrm{Var}(\widehat{v}_i - \widehat{v}_j) = u_i^2 + u_j^2 - 2\,\mathrm{Cov}(\widehat{v}_i, \widehat{v}_j).$$

The independent-route formula is a special case. A route absent an assumption is useful only if it is genuinely insensitive to the associated error and has adequate resolution. Agreement is evidence conditional on these distinctions, not a universal proof of physical truth.

14 Why mature RG dismantles coherence as a foundation

The phrase must be stated carefully. RG does not refute wave interference, quantum coherence, synchronization, or stable statistical dependence. It dismantles the stronger inference that coherence itself guarantees an identifiable reality or a uniquely determined geometry.

14.1 Coherence is not sufficient

Take two perfectly phase-aligned contributions with observation

$$Y = a + b + \epsilon.$$

The phase order parameter is one, yet (a, b) and $(a + \delta, b - \delta)$ have the same law for fixed noise. Perfect phase agreement leaves an exact identification fibre. A stable repeated return can be precisely estimated while its decomposition remains unresolved.

An exactly coincident pair of echoes gives the same lesson at waveform level:

$$A_1 p(t - \tau) + A_2 p(t - \tau) = (A_1 + A_2) p(t - \tau).$$

No detector can distinguish those descriptions from that waveform alone under the same noise law. Coherence can make the signal stronger without making its latent origin unique.

14.2 Coherence is not necessary

Take separately observed labelled channels

$$Y_1 = a + \epsilon_1, \qquad Y_2 = -b + \epsilon_2.$$

Their phases may be 0 and π, giving zero aggregate phase order, while the amplitude map has full rank and identifies (a, b) under the declared law. The failure of one aggregate coherence statistic does not imply failure of every task.

These examples concern an explicit phase-order notion, not every possible use of the word coherence. Their force is that no unspecified universal coherence criterion can replace a declared observation map and task.

14.3 A scalar cannot carry the entire certificate

A single score can hide which of the following failed: units, path identity, scale, rank, conditioning, global separation, model adequacy, projectability, or physical calibration. The archive's "Modulation Compatibility Index" section makes the issue visible. It defines

$$\vec{K} = \nabla S / \mathcal{S}_*, \qquad \mu = \frac{|\vec{K}|\Omega}{\Theta \mathcal{S}_*},$$

then calls μ dimensionless while assigning $\vec{K}$ momentum units. For spatial derivatives, the normalized gradient actually has inverse-length units. Even if the printed momentum units were used instead, the displayed ratio would have units $\mathrm{m}^{-1}\mathrm{s}^{-1}$, not one. The intended cross-domain compatibility idea is historical evidence; the printed dimensional closure is not a valid invariant theorem [2].

Mature RG cannot fix this by declaring a better universal score. It has to type each map, declare each input, and retain the separate failure modes. That is a methodological change, not merely a corrected numerical coefficient.

14.4 What the childhood intuition got right

The original intuition correctly kept returning to the relation between an event and its confirmation, and to the possibility that changing a protocol changes what can be resolved. Its overreach was to let recognizable persistence stand in for justified identification. The mature account does not need to ridicule that beginning. It explains why it was fertile and why it could not remain the foundation.

Proposed retrospective authorial formulation. I began by believing in coherence. The question that survived was not whether a response could be made to hold together, but what its holding together entitled me to claim. RG emerged when that entitlement became the object of the mathematics. This paragraph is a proposed first-person formulation of the retrospective thesis, not a quotation recovered from childhood notes.

15 Admissibility gates: why geometry had to survive changes of protocol

The mature programme's rank stability, bounded conditioning, monotonicity under coarse-graining, and transport/composability gates make explicit what a scalar coherence criterion could not certify. Rank tests infinitesimal sensitivity; conditioning tests whether inversion amplifies finite error; coarse-graining tests whether a claimed distinction was manufactured by discarding data; composability tests whether the construction remains meaningful when observation or transport steps are chained. These are separate obligations, not four names for the same survival score [18, 11, 13].

A finite statistical calculation makes the coarse-graining bridge precise. Let $p_\theta(y) > 0$ be a differentiable finite model and let a parameter-independent stochastic map produce Z from Y. The score of the coarsened experiment is

$$s_Z(z) = \mathbb{E}_\theta[s_Y(Y) \mid Z = z].$$

Conditional variance then gives

$$I_Y - I_Z = \mathbb{E}_\theta[\mathrm{Cov}_\theta(s_Y \mid Z)] \succeq 0.$$

Thus coarse-graining cannot increase Fisher information under these assumptions. Equality means that the score is already measurable through the retained observation at that parameter; it is not by itself a global reconstruction theorem. A data-processing axiom is therefore a substantive selection requirement, not a consequence of saying that a response is coherent.

The earlier coherence-density and clock-rate proposals also belong to this bridge history. They attempted to connect response survival to temporal structure. Mature RG cannot promote a coherence-dependent clock formula to physical proper time without specifying the clock observable, its calibration, the dynamics, and the relevant equivalence. A scalar density can remain a model feature while losing its status as an automatic clock or universal metric source [2, 7, 18].

Finally, a bounded condition number on one chart does not prove a uniform bound over every admissible state. Near rank-changing strata, a regular local quotient and its metric description can cease to apply. The framework must report the boundary of its regular regime rather than interpret every failure as a single coherence rupture.

16 The uniqueness crisis and the repaired ladder

The strongest origin story would say that the wall already forced the whole mature geometry. The repaired ladder rejects that shortcut. At least four statements must be separated:

1. an object can be constructed from declared data;
2. it is constant on observational fibres and therefore descends;
3. it is natural under a specified class of representation changes;
4. it is uniquely characterized among all candidates satisfying stated axioms.

None of the first three automatically implies the fourth. For example, on a Bernoulli parameter interval both dp^2 and $dp^2/[p(1-p)]$ are invariant under outcome exchange, but they are not constant

multiples. This does not refute stronger metric-characterization theorems with additional axioms. It refutes the claim that this invariance alone selects the metric.

The mature theorem must therefore have the form

$$\text{declared inputs} + \text{candidate class} + \text{hypotheses} \Longrightarrow \text{conclusion up to stated equivalence.}$$

Where uniqueness fails, a classification, obstruction, or correspondence may be the correct result. A failed inevitability claim need not erase a useful construction. Conversely, useful computations do not rescue an unproved universal uniqueness claim [18, 11, 12].

The real hunt for RG is reconstructed here as a sequence of such separations. It begins when a coherent inverse ceases to count as a sufficient explanation and continues when even a canonical quotient ceases to count as a sufficient geometry.

17 The full dependency roadmap

Table 1 collects the bridges. "Needed" means needed to justify the corresponding claim in this reconstructed programme, not that every mathematician must discover these named CTMT devices in this order.

Table 1: From delayed return to mature RG: bridge, achievement, and remaining gap.

	bridge	what it made possible	what it did not establish
1	Wall-tap ordering	Event, interval, return, protocol dependence	Calibrated propagation time
2	Green response	Linear causal impulse-to-response model	Unique inverse or derived medium
3	Spectral reconstruction	Envelope, phase, frequency and response structure	Delay from power alone
4	RMI assembly	Common language for operational kernel families	Universal physical generator
5	Sensing map and RIP	Computable inverse with noise and regularization	Observational uniqueness
6	MKP and ICM	Recursive prediction and information-survival questions	Calibrated universal coherence score
7	Primitive extraction and anchors	Measurement links, units and scale constraints	Independent emergence of imported constants
8	FMC and TUCF	Spatial/temporal composition and uncertainty windows	Task sufficiency of compressed features
9	Terror deformation	Rupture, shocks and hostile alternatives	One universal collapse mechanism
10	Redundancy and rigidity	Cross-route weighting and phase-selective filtering	Unique latent interpretation
11	Fisher and CRSC	Directional sensitivity and stability diagnostics	Causal null cone from statistical degeneracy
12	Information trigonometry	Statistical lengths and model-based angles	Physical metres from flatness
13	Resolved/unresolved elimination	Conditional covariance and memory kernels	Feedback from covariance alone
14	Exact law quotient	Canonical observational classes and task descent	A selected metric or dynamics
15	Gauge completeness	Separation of known symmetries from full fibres	Completeness from soundness alone
16	Saturation and presentation	Relative separation and observable portability	Arbitrary latent reconstruction
17	Projectability and transport	Representative-independent observable evolution	A canonical latent lift without extra data

	bridge	what it made possible	what it did not establish
18	Cone and characteristic structure	Conditional causal/conformal geometry	Signature or scale from Fisher alone
19	Clock, medium and length anchors	Typed duration, path length and calibrated distance	Universal equivalence of radar and rods
20	Consilience and held-out tests	Agreement across vulnerable routes	Detection of shared bias or all alternatives
21	Repaired uniqueness discipline	Explicit candidate classes and equivalences	Unconditional inevitability of all geometry

18 What the computations do and do not add to this history

The old Python is evidence of what was operationalized, not evidence that every surrounding physical claim was established. The current audit inventories all 31 preformatted blocks in the supplied RMI–Terror extract; 26 parse as Python. Ten selected blocks were invoked independently without repairing missing notebook state. Nine returned exit code zero; one failed because it refers to an undefined variable. One of the zero-exit blocks only loads function definitions and does not exercise them. Exit code zero is not a scientific acceptance test.

The reruns preserve important discrepancies. The basic inversion example divides its residual statistic by zero degrees of freedom. A related example reports an undefined reduced statistic explicitly. The temporal and joint spatial–temporal demonstrations print large diagnostic quantities rather than automatically confirming their model. These outcomes are retained in the logs. They do not prove that every possible version of the programme fails; they show that the supplied demonstrations cannot bear stronger claims than their actual outputs.

The new companion `rg_origin_checks.py` executes fourteen mathematical witnesses: coherence without identification, identification without aggregate phase coherence, power-spectrum shift ambiguity, regularizer dependence, local/global separation, statistical versus causal nullity, flat Fisher scale ambiguity, shared route bias, covariance without feedback, exact elimination, lift ambiguity, metric nonselection, conformal scale ambiguity, and nontransitive thresholds. All fourteen pass their specified numerical or analytic checks. They are illustrative counterexamples and identities, not empirical measurements or a validation of the entire RG ladder.

The supplied final wall-tap print is a separate synthetic waveform study. This origin paper does not rerun that experiment or turn its reported results into historical observations. Its role here is to show how the mature programme returns to the original question with calibration, uncertainty, held-out evaluation, and explicit unresolved alternatives [19].

19 Conclusion: the wall remained; the criterion changed

The Green kernel was an appropriate first language because it wrote the easiest part of the problem: an impulse produces a response after a delay. RMI enlarged the response language. Terror made failure and survival explicit. The intermediate work introduced sensing, inversion, anchors, covariance, directional information, compression, and cross-route comparison. Those were genuine bridges, but none licensed the conclusion that a coherent construction was the only geometry compatible with the record.

RG emerged through the repeated removal of that inference. A response kernel became an observation problem. A stable estimate became a fibre of alternatives. A local rank became one diagnostic among global separation and finite-resolution tests. A null direction became a typed mathematical object rather than an automatic causal channel. A quotient became a foundation for further conditional structure, not a machine that selected every metric, connection, and physical scale by itself.

The mature framework does not say that coherence never matters. It says that coherence must state its observable, model, and task like any other property. It cannot certify its own ontology.

What survives from the childhood beginning is the insistence that an event and its confirmation are not the same thing, and that the interval between them is a question to investigate rather than a permission to assume the answer.

A Source ledger and historical limits

The source package records SHA-256 identities and the archive's heading inventory. The following locators make the narrative inspectable without treating file modification times as discovery dates.

source	locator	use in this paper
Author's account	Current authorial statements on Green-first representation and childhood coherence	Recollection, not independent dating
CTMT HTML	Green Kernel (linear propagation); Green Kernel (Spectral); Wall-Tap Delays and Primitive Terror Calculus	Response model and autobiographical source
RMI–Terror extract	Recursive Modulation Impulse; Terror Kernel; Modulation-Derived Acceleration	Assembly, inversion, recursion, rupture, code and unit audit
CTMT HTML	Amplitude potential seed and CRSC; Collapse as Projection–Rupture	Null projection and directional stability semi-steps
Trigonometry papers	Fisher foundation, stationary-phase scaling, classical-limit claims	Statistical geometry and its physical-scale gap
Origins and Elimination	Null controls, Schur correction, elimination kernel	Coupling types and conditional memory
Reconstruction and Saturation	Observable completeness and relative separation	Fibre separation versus reconstruction
Assembly and repaired RG	Kernel groupoid, gauge completeness, projectability, cone descent	Mature structural distinctions
Canonical Transport	Gauge shear no-go and declared covariance	Lift nonuniqueness and conditional selection
Uncertainty Geometry	Covariance and task geometry	Error structure rather than scalar confidence
Physical Distance	Calibration and identification hypotheses	Statistical length versus metres
Presentation and Thresholds	Exact kernel and finite decision layer	Repair of threshold equivalence
Crossing into Physics	Route assumptions, clusters, common residue	Consilience and shared-bias limitation
Final wall-tap print	Formal model, synthetic evidence and non-claims	Modern return to the motivating observation

The ledger is a conceptual reconstruction of the supplied corpus, not a claim that every line of every CTMT document has been validated. The full extract was structurally inventoried, including its third acceleration section; the code audit is scoped explicitly. No archival statement of "first" is inferred solely from the current HTML ordering.

B Three old formulas that require typing before interpretation

B.1 Action, phase, and momentum

If S has action units, ∇S has momentum units for spatial differentiation, whereas $\nabla S/S_*$ has inverse-length units. A normalized phase gradient is not simultaneously an unnormalized momentum. The action divisor must also be reconciled with cycle versus angular frequency. The origin paper records these differences rather than proposing a replacement universal modulation index.

B.2 Latent and observed null spaces

For $J : T_xX \to T_{F(x)}Y$, $\ker J$ is latent and $\ker J^*$ is observation-normal after metrics are specified. For positive-definite observation covariance,

$$I = J^\top \Sigma^{-1} J, \qquad \ker I = \ker J.$$

An observation-space Gram matrix and a latent Fisher matrix can have different dimensions. Their nonzero singular-value information is related, but their kernels are not interchangeable. This is the exact repair needed when an old pipeline stores its regression Jacobian transposed.

B.3 Curvature, degeneracy, and acceleration

A stationary-phase Hessian, a Fisher information matrix, and a Riemann curvature tensor are different objects even if the archive uses similar letters or language. A derivative of velocity with respect to an abstract curvature coordinate is not a physical acceleration without a clock map. These distinctions prevent an attractive geometric vocabulary from doing the work of missing hypotheses.

C Entropy and the unresolved fibre

The entropy quotient is another semi-step that survives only with its measure assumptions visible. In a finite probability model,

$$H(X) = H(Q) + \mathbb{E}_Q H(X \mid Q)$$

separates uncertainty between observable classes from uncertainty within them. A maximum-entropy completion chooses a distribution relative to declared constraints and a reference measure; it does not make hidden alternatives observable. An unrestricted differential-entropy identity is not inferred from the finite formula. This is the same general reversal: unresolved structure is recorded, not removed by a coherence preference.

D Code and witness ledger

Block indices below are zero-based positions among the 31 `pre` elements in the supplied extract. Extraction is mechanical and the original source hash is recorded in `rg_origin_archive_audit.json`. Some blocks are pseudocode, output, or another language; Python syntax failure alone is not evidence of a mathematical error.

block	role	independent rerun status
4	Symbolic/numeric impulse-kernel derivation	completed; illustrative anchors
7	Basic weighted inversion demonstration	completed; zero-degree-of-freedom warning
8	Kernel-origin/Terror demonstration	completed; undefined reduced statistic retained
9	Temporal projection demonstration	completed; diagnostic output retained
10	Joint spatial/temporal demonstration	completed; diagnostic output retained
21	Coherence/rupture/Lyapunov illustration	completed; toy diagnostic output
25	Redundancy/rigidity ensemble implementation	completed; toy-model output
28	CRSC illustration	completed; toy diagnostic output
29	Dependent notebook fragment	failed: undefined variable
30	Projection–rupture analysis functions	definitions loaded; functions not invoked

witness	mathematical point	result
OC01	Perfect phase alignment can leave an exact latent fibre	pass
OC02	Zero aggregate phase order can coexist with labelled identification	pass
OC03	Equal power spectra can hide a time shift	pass
OC04	Different regularizers select different representatives	pass
OC05	Differential rank and global injectivity differ	pass
OC06	Fisher radical and Lorentz null cone differ	pass
OC07	Flat Fisher geometry does not fix a metre scale	pass
OC08	Pairwise route agreement does not detect common bias	pass
OC09	Cross covariance need not imply feedback memory	pass
OC10	Declared elimination produces its memory kernel	pass
OC11	Law-preserving shears change latent lifts	pass
OC12	Outcome-exchange invariance does not select a metric	pass
OC13	A cone leaves a positive scale freedom	pass
OC14	Threshold closeness need not be transitive	pass

OC05 is an analytic witness recorded by the script, not a numerical proof of global injectivity. The other checks similarly have the scope stated in their source. None is an empirical acoustic test.

E Reproduction and editorial accountability

The companion bundle contains this LaTeX source, the compiled PDF, the witness script and JSON, archive audit, extracted code blocks and selected execution logs, and a source-identity manifest. The main paper is standalone; the audit material allows inspection of the older implementation without silently rewriting it into mature RG.

```
python rg_origin_checks.py --out .
pdflatex RG_Origin_From_the_Wall.tex
pdflatex RG_Origin_From_the_Wall.tex
```

The archival audit script additionally requires BeautifulSoup and the supplied `RMI + Terror Kernel.txt`; selected blocks require their own imported scientific libraries. Running them as separate programs intentionally does not supply hidden notebook variables. The bundle does not claim an independent preregistration, empirical dataset, or historical timestamp beyond the source identities it records.

References

[1] M. Rada, authorial recollections in the accompanying research conversation. Green-first representation and childhood belief in coherence are attributed to these statements; explanatory interpretation is identified in the text.

[2] M. Rada, *CTMT HTML archive*, supplied `index.html`. In particular the Green-kernel, wall-tap, CRSC, and projection–rupture sections identified in Appendix A.

[3] M. Rada, *RMI + Terror Kernel*, supplied HTML-formatted text extract. Three principal sections and 31 preformatted blocks; source identity and block inventory accompany this paper.

[4] M. Rada, *From Wall-Tap Delay to Operational Resolution Geometry*, supplied *RG – Origins* manuscript.

[5] M. Rada, *From a Child's Wall-Tap Observation to Resolution Geometry*, supplied *RG – Wall-Tap Delay* manuscript. Earlier formulations are not treated as superseding the repaired exact-law definitions.

[6] M. Rada, *CTMT Trigonometry: Information–Geometric Distances and the Classical Limit*, supplied manuscript.

[7] M. Rada, *Information–Geometric Trigonometry*, supplied manuscript.

[8] M. Rada, *RG – Elimination*, supplied manuscript.

[9] M. Rada, *RG – Reconstruction Identifiability*, supplied manuscript.

[10] M. Rada, *RG – Saturation*, supplied manuscript.

[11] M. Rada, *Resolution Geometry as an Observable Orbit–Cone Structure*, supplied *Assembly of Resolution Geometry* manuscript.

[12] M. Rada, *RG – Canonical Transport*, supplied manuscript.

[13] M. Rada, *RG – Uncertainty Geometry*, supplied manuscript.

[14] M. Rada, *RG – Residual Characteristic Geometry*, supplied manuscript.

[15] M. Rada, *RG – Physical Distance*, supplied manuscript.

[16] M. Rada, *RG – Presentation and Thresholds*, supplied manuscript.

[17] M. Rada, *RG – Crossing into Physics*, supplied manuscript.

[18] M. Rada, *Resolution Geometry*, supplied repaired print edition. Used for the distinction among exact quotient, descent, selection, conditional rigidity, and physical calibration.

[19] M. Rada, *Wall–Tap Delay: Calibrated Duration, Acoustic Distance, and Identifiability*, supplied final computational print. Synthetic results are not recast as childhood measurements.